

B. Tech 2nd Semester Sessional Examination-II
Subject-Mathematics-II, Branch- CSE (Sec-A), Total Marks-30, Date-29.04.2025

Answer the following questions:

1. Solve (**any three**):

5x3=15

- a) $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} = e^{2x}$
- b) $\frac{d^2y}{dx^2} + y = \cos 2x$
- c) $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} - 3y = x$
- d) $x^2 \frac{d^2y}{dx^2} - 3x \frac{dy}{dx} = 0$
- e) $\frac{d^2y}{dx^2} + 4y = \sec 2x$ (use method of variation of parameters)
- f) $\frac{d^2y}{dx^2} - y = 0$ (in power series)

2. Using Cauchy Integral Formula, evaluate **any three** of the integrals:

5x3=15

- (a) $\int_C \frac{zdz}{(9-z^2)(z+i)}$ where C is $|z| = 2$
- (b) $\int_C \frac{(8z-3)dz}{z^2-z}$ where C is $|z| = 2$
- (c) $\int_C \frac{z^2dz}{z^2+4}$ where C is $|z+2i| = 1$
- (d) $\int_C \frac{10dz}{(z+i)^4}$ where C is $|z+i| = 1$
- (e) $\int_C \left(z + \frac{1}{z^2}\right) dz$ where C is $|z| = 2$